Uncertainty Quantification in Machine Learning

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Chapter 1

Assignment 1 Done 2026 September 24 Due 2026 September 29

Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 1

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$ and a given model $\hat{f} : \mathcal{X} \to \mathbb{R}$. Construct a score function $s(x, y)$ that would yield intervals of the form $\mathcal{C}(x) = [\hat{f}(x) - \gamma, \hat{f}(x) + 2\gamma]$ for some $\gamma > 0$.

Solution Put

$$s(x,y) = \begin{cases} 2\left|\hat{f}(x) - y\right| & y < \hat{f}(x) \\ \left|\hat{f}(x) - y\right| & y \geqslant \hat{f}(x). \end{cases}$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leqslant \hat{q}\}$, we will have the desired interval with $\gamma = \frac{\hat{q}}{2}$. ////

Exercise 2

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}^d$ and a given model $\hat{f} : \mathcal{X} \to \mathbb{R}^d$ whose coordinates are $\hat{f}(x) = (\hat{f}_1(x), \ldots, \hat{f}_d(x))$. Construct a score function $s(x, y)$ that would yield intervals of the form

$$\mathcal{C}(x) = \prod_{j=1}^{d} [\hat{f}_j(x) - c, \hat{f}_j(x) + c]$$

for a $c > 0$.

Solution Put

$$s(x,y) = \max_{j=1,\ldots,d} \left|\hat{f}_j(x) - y_j\right|.$$

Then, with $\hat{q}$ being a quantile of the scores on the validation set and $\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leqslant \hat{q}\}$, we will have the desired interval with $c = \hat{q}$. ////

Exercise 3

Consider split conformal prediction with $\mathcal{Y} = \mathbb{R}$, two given models $\hat{f}_i : \mathcal{X} \to \mathbb{R}$ for $i = 1, 2$, and the score function

$$s(x, y) = \min_{i=1,2} \left| y - \hat{f}_i(x) \right|.$$

Derive an explicit expression for $\mathcal{C}(x)$ in terms of $\hat{f}_1(x)$, $\hat{f}_2(x)$, and the quantile $\hat{q}$ of the scores on the validation set. What is the shape of the prediction when $\hat{q}$ is small or when it is large? Briefly explain why this might be attractive if the conditional distribution of Y given $X = x$ is bimodal.

Solution By definition, it is immediately clear that

$$\mathcal{C}(x) = \{y \in \mathcal{Y} \mid s(x, y) \leqslant \hat{q}\} = [\hat{f}_1(x) - \hat{q}, \hat{f}_1(x) + \hat{q}] \cup [\hat{f}_2(x) - \hat{q}, \hat{f}_2(x) + \hat{q}].$$

If $\hat{f}_1(x) = \hat{f}_2(x)$, then $\mathcal{C}(x)$ is an interval of length $2\hat{q}$ centered at $\hat{f}_1(x) = \hat{f}_2(x)$. If we assume that the two models are different at x, that is, $\hat{f}_1(x) \neq \hat{f}_2(x)$, then $\mathcal{C}(x)$ is the union of two intervals when $\hat{q}$ is small enough, and it is a single interval when $\hat{q}$ is large enough. If the conditional distribution of Y given $X = x$ is bimodal, then the score might be of interest because $\hat{f}_1(x)$ and $\hat{f}_2(x)$ might capture the two modes of the distribution and the predicted interval will be centred at both modes. ////

Exercise 4

Fix $n > 2$. Construct a random vector $(X_1, \ldots, X_n)$ defined on some probability space such that (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \ldots, X_n)$ is not exchangeable.

Solution Let (X, Y) be a jointly normal random vector. It is exchangeable if and only if $\mathbb{E}[X] = \mathbb{E}[Y]$ and $\text{Var}(X) = \text{Var}(Y)$. With this in mind, let $(X_1, \ldots, X_n)$ be a jointly normal random vector such that $\mathbb{E}[X_i] = \mathbb{E}[X_j]$ and $\text{Var}(X_i) = \text{Var}(X_j)$ for all $i \neq j$ but $\text{Cov}(X_1, X_2) \neq \text{Cov}(X_1, X_3)$. Then (X_i, X_j) is exchangeable for all $i \neq j$ but $(X_1, \ldots, X_n)$ is not exchangeable. For a concrete example, say $\mathbb{E}[X_i] = 0$ and $\text{Var}(X_i) = 1$ for all i, $\text{Cov}(X_1, X_2) = -1$, and all other covariances are zero. ////

Exercise 5

Fix $n \geqslant 1$, $\tau \in (0, 1)$. For any vector $v \in \mathbb{R}^n$, define $\text{Quantile}(v; \tau) = v_{(\lceil \tau n \rceil)}$, where $v_{(1)} \leqslant v_{(2)} \leqslant \ldots \leqslant v_{(n)}$ are the order statistics of v. Under what condition is it true that, for all $v \in \mathbb{R}^n$, $\text{Quantile}(v; 1 - \tau) = -\text{Quantile}(-v; \tau)$?

Solution The order statistics of $-v$ are given by $(-v)_{(i)} = -v_{(n-i+1)}$ for $i = 1, \ldots, n$. Therefore, $\text{Quantile}(v; 1-\tau) = -\text{Quantile}(-v; \tau)$ holds for a vector $v \in \mathbb{R}^n$ if and only if

$$v_{(\lceil (1-\tau)n \rceil)} = -((-v)_{(\lceil \tau n \rceil)}) = -(-v_{(n - \lceil \tau n \rceil + 1)}) = v_{(n - \lceil \tau n \rceil + 1)}.$$

To make this work for all $v \in \mathbb{R}^n$, the condition we must pose is that

$$\lceil (1 - \tau)n \rceil = n - \lceil \tau n \rceil + 1.$$

This can happen for instance when $\tau = 1/2$ and n is odd. ////

Exercise 6

For $z \in \mathbb{R}^n$, write $\hat{F}_z(v) = \frac{1}{n}\sum_{i=1}^n \chi_{\{z_i \leq v\}}$ for the empirical cumulative distribution function of z with $\hat{F}_z(-\infty) = 0$ and $\hat{F}_z(+\infty) = 1$. Write

$$\text{Quantile}(z;\tau) = \inf\left\{v \in \mathbb{R} \mid \hat{F}_z(v) \geqslant \tau\right\}$$

for the empirical τ-quantile of z for any $\tau \in [0,1]$. Consequently, $\text{Quantile}(z;0) = -\infty$.

Prove that, for any $z \in \mathbb{R}^n$, the following statements are true:

1. $\hat{F}_z(v) = \sup\{\tau \in [0,1] \mid \text{Quantile}(z;\tau) \leqslant v\}$ for all $v \in \mathbb{R}$.
2. $\text{Quantile}(z; \hat{F}_z(v)) \leqslant v$ for all $v \in \mathbb{R}$.
3. $\hat{F}_z(\text{Quantile}(z;\tau)) \geqslant \tau$ for all $\tau \in [0,1]$.
4. If z has no repeated values, then $\hat{F}_z(\text{Quantile}(z;\tau)) = \frac{\lceil n\tau \rceil}{n}$ for all $\tau \in [0,1]$.

Solution Let $z \in \mathbb{R}^n$ be given and fixed. Let us inrtoduce a notation S_τ for the set of all $v \in \mathbb{R}$ such that $\hat{F}_z(v) \geqslant \tau$. Then, by definition, $\text{Quantile}(z;\tau) = \inf S_\tau$. All four statements depend on the following lemma:

Lemma 1. If $v \in \mathbb{R}$ and $\tau \in [0,1]$, then $\text{Quantile}(z;\tau) \leqslant v$ if and only if $\tau \leqslant \hat{F}_z(v)$.

Proof. If $\tau \leqslant \hat{F}_z(v)$, then $v \in S_\tau$ and $v \geqslant \inf S_\tau = \text{Quantile}(z;\tau)$.

Let us focus on the converse. Suppose that $\text{Quantile}(z;\tau) \leqslant v$. For simplicity, put $q = \text{Quantile}(z;\tau)$. If $\tau = 0$, then $\hat{F}_z(v) \geqslant 0 = \tau$ holds trivially. Assume from now on that $\tau > 0$. Then q is finite number. Choose $\delta > 0$ small enough such that no points z_i lies between q and $q + \delta$, which is always doable because z has only finitely many points. Since q is the infimum, choose $s \in S_\tau$ such that $q \leqslant s < q + \delta$. Then, we arrive at

$$\hat{F}_z(v) \geqslant \hat{F}_z(q) = \hat{F}_z(s) \geqslant \tau$$

by applying in the first inequality the monotonicity of $\hat{F}_z$ and in the second equality the fact that no points z_i lies between q and $q + \delta$. The lemma is proved. □

We now prove the four statements. To prove the first statement, let $v \in \mathbb{R}$ be given and we have, by the lemma:

$$\{\tau \in [0,1] \mid \text{Quantile}(z;\tau) \leqslant v\} = \left\{\tau \in [0,1] \mid \tau \leqslant \hat{F}_z(v)\right\} = [0, \hat{F}_z(v)]$$

and the suprenum of the interval is just $\hat{F}_z(v)$. The second and the third statements are already included in the lemma. The fourth statement is trivial given the special assumption that z has no repeated values.////

Exercise 7

Consider the conformal prediction with $\mathcal{Y} = \mathbb{R}$ with a given model $\hat{f} : \mathcal{X} \to \mathcal{Y}$ and calibration points (X_1, Y_1),..., (X_n, Y_n). Put

$$\hat{q}_L = \text{Quantile}\left(\left\{\hat{f}(X_i) - Y_i\right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right)\left(1 + \frac{1}{n}\right)\right),$$

$$\hat{q}_R = \text{Quantile}\left(Y_i - \left\{\hat{f}(X_i)\right\}_{i=1}^n ; \left(1 - \frac{\alpha}{2}\right)\left(1 + \frac{1}{n}\right)\right),$$

and

$$\mathcal{C}(X_{n+1}) = \left[\hat{f}(X_{n+1}) - \hat{q}_L, \hat{f}(X_{n+1}) + \hat{q}_R\right].$$

Prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \geqslant 1-\alpha$$

by showing both

$$P\left\{Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L\right\} \leqslant \frac{\alpha}{2}$$

and

$$P\left\{Y_{n+1} > \hat{f}(X_{n+1}) + \hat{q}_R\right\} \leqslant \frac{\alpha}{2},$$

assuming that $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$ is exchangeable.

Solution We have

$$P\left\{Y_{n+1} < \hat{f}(X_{n+1}) - \hat{q}_L\right\} = P\left\{\hat{f}(X_{n+1}) - Y_{n+1} > \hat{q}_L\right\} = 1 - P\left\{\hat{f}(X_{n+1}) - Y_{n+1} \leqslant \hat{q}_L\right\} \leqslant 1 - \left(1 - \frac{\alpha}{2}\right) = \frac{\alpha}{2}$$

where the last inequality uses Theorem 3.2 in the textbook with score $S_i = \hat{f}(X_i) - Y_i$. This is legitimate because the data is assumed to be exchangeable and the score is trivially symmetric in case of split conformal prediction. The right hand side inequality is proved in exactly the same way but with score $S_i = Y_i - \hat{f}(X_i)$. Combine the two, we have the desired marginal coverage. The proof is complete. ////

Exercise 8

Consider the full conformal prediction with $(X_1, Y_1), \cdots, (X_{n+1}, Y_{n+1})$ being exchangeable and a symmetric score function s, prove that

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \leqslant 1 - \alpha + \mathbb{E}\left[\frac{1}{n+1}\sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}}\right]$$

where

$$\hat{q} = \text{Quantile}\left(S_1, \ldots, S_{n+1}; 1-\alpha\right).$$

Give an example showing that this bound is tighter than

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} \leqslant \frac{\lceil (1-\alpha)(n+1) \rceil}{n+1} + \epsilon_{\text{tie}}$$

where ϵ_{tie} is the probability of S_{n+1} being equal to some S_j for $j \in \{1, \ldots, n\}$.

Solution As already discussed in the lecture and in the textbook, $Y_{n+1} \in \mathcal{C}(X_{n+1})$ if and only if $S_{n+1} \leqslant \hat{q}$ and so

$$P\left\{Y_{n+1} \in \mathcal{C}(X_{n+1})\right\} = P\left\{S_{n+1} \leqslant \hat{q}\right\} = P\left\{S_{n+1} < \hat{q}\right\} + P\left\{S_{n+1} = \hat{q}\right\}.$$

Given that $(S_1, \ldots, S_n, S_{n+1})$ is exchangeable, $P\left\{S_{n+1} = \hat{q}\right\} = P\left\{S_i = \hat{q}\right\}$ for all $i \in \{1, \ldots, n\}$ and

$$P\left\{S_{n+1} = \hat{q}\right\} = \frac{1}{n+1}\sum_{i=1}^{n+1} P\left\{S_i = \hat{q}\right\} = \mathbb{E}\left[\frac{1}{n+1}\sum_{i=1}^{n+1} \chi_{\{S_i = \hat{q}\}}\right].$$

The only thing left is to show that $P\{S_{n+1} < \hat{q}\} \leqslant 1-\alpha$. To do so, first observe that $S_{n+1} < \hat{q}$ if and only if $S_{n+1} < S_{(k)}$, where $k = \lceil (1-\alpha)(n+1) \rceil$. Thus:

$$P\{S_{n+1} < \hat{q}\} = P\left\{S_{n+1} < S_{(k)}\right\} \leqslant \frac{\lceil (1-\alpha)(n+1) \rceil - 1}{n+1} = \frac{\lfloor (1-\alpha)(n+1) \rfloor}{n+1} \leqslant \frac{(1-\alpha)(n+1)}{n+1} = 1-\alpha.$$

This completes the proof.

Finally, in case when the random variables S_i's are all continuous random variables, $\epsilon_{\text{tie}} = P\{S_{n+1} = \hat{q}\} = 0$ and evidently the bound that is worked out in this exercise is better. In case when the random variables S_i's are discrete and can only take very limited values, it may happen that $S_1, \ldots, S_n$ has taken all possible values and $\epsilon_{\text{tie}} = 1$ so that the bound worked out here is really the only available bound. ////

Exercise 9

Given data set $(X_1, Y_1), \ldots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ where $\mathcal{Y} = \mathbb{R}$, consider split conformal prediction with two given models $\hat{f}_i : \mathcal{X} \to \mathbb{R}$, the score functions

$$s_i(x, y) = \left| y - \hat{f}_i(x) \right|,$$

and two split conformal prediction sets

$$\mathcal{C}_i(X_{n+1}) = \{y \in \mathcal{Y} \mid s_i(X_{n+1}, y) \leqslant \hat{q}_i\},$$

where

$$\hat{q}_i = \text{quantile}\left\{ \{s_i(X_j, Y_j)\}_{j=1}^n ; (1-\alpha)(1+\frac{1}{n}) \right\}.$$

Put

$$L_i = \frac{1}{n} \sum_{j=1}^{n} (Y_j - \hat{f}_i(X_j))^2$$

and $I = \arg\min_{i \in \{1,2\}} L_i$. Explain why

$$P\{Y_{n+1} \in \mathcal{C}_I(X_{n+1})\} \geqslant 1-\alpha$$

may not hold even if we assume that $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$ are exchangeable.

Solution In one sentence, the reason is that even if we assume that the three random vectors $(X_1, Y_1), \ldots, (X_{n+1}, Y_{n+1})$, $s_1(X_1, Y_1), \ldots, s_1(X_{n+1}, Y_{n+1})$, and $s_2(X_1, Y_1), \ldots, s_2(X_{n+1}, Y_{n+1})$ are all exchangeable, in general it is still not true that $s_I(X_1, Y_1), \ldots, s_I(X_{n+1}, Y_{n+1})$ is exchangeable and thus the theorem guaranteeing the marginal coverage does not in general apply.

To show that things can indeed fail, we give a simple counterexample. Put $\mathcal{X} = \mathbb{R}^2$, $n = 1$, $\alpha = \frac{1}{2}$, $X_j = (X_j^{(1)}, X_j^{(2)})$, X_j follows $N_2(0, I_2)$ distribution, $Y_j = X_j^{(1)} + X_j^{(2)}$, $\hat{f}_1(x) = x^{(1)}$ and $\hat{f}_2(x) = x^{(2)}$. It is clear that, for $i = 1, 2$, $\hat{q}_i = s_i(X_1, Y_1)$, $s_i(X_j, Y_j) = \left| X_j^{(2-i)} \right|$, and $L_i = (X_1^{(2-i)})^2$. Thus,

$$\begin{aligned}
&P\{s_I(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\} \\
&= P\{s_1(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 < L_2\} \\
&+ P\{s_2(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1)), L_1 > L_2\} \\
&= P\left\{ \left|X_2^{(2)}\right| \leqslant \left|X_1^{(2)}\right| \leqslant \left|X_1^{(1)}\right| \right\} + P\left\{ \left|X_2^{(1)}\right| \leqslant \left|X_1^{(1)}\right| \leqslant \left|X_1^{(2)}\right| \right\} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}
\end{aligned}$$

and

$$P\left\{Y_2 \in \mathcal{C}_I(X_2)\right\} = P\left\{s_I(X_2, Y_2) \leqslant \min(s_1(X_1, Y_1), s_2(X_1, Y_1))\right\} = \frac{1}{3} < \frac{1}{2} = 1 - \alpha.$$

This completes the counterexample. ////

Chapter 2

Assignment 2 Done 2026 October 16 Due 2026 October 18

Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 1

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+1}, Y_{n+1}) \in \mathcal{X} \times \mathcal{Y}$ and the conformal risk control with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. Instead of what is in Chapter 10 of the book, we define, for all $\lambda \geqslant 0$:

$$\mathcal{C}_\lambda(x) = \{y \in \mathcal{Y} \mid s(x, y) < \lambda\}$$

and

$$L(y, \mathcal{C}_\lambda(x)) = \chi_{\{y \notin \mathcal{C}_\lambda(x)\}}.$$

Show that, with this definition, $L(y, \mathcal{C}_\lambda(x))$ is no longer always right continuous for every fixed $(x, y) \in \mathcal{X} \times \mathcal{Y}$. Next, given $\alpha \in (0, 1)$, give a concrete expression for

$$\hat{\lambda} = \inf \left\{ \lambda \geqslant 0 \mid \hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n} \right\}$$

in terms of the data, where

$$\hat{R}(\lambda) = \frac{1}{n} \sum_{i=1}^{n} L(Y_i, \mathcal{C}_\lambda(X_i)).$$

Show that, if we assume that $(S_1, \ldots, S_{n+1})$ is exchangeable and that the S_i's almost surely has no tie, then

$$P\left\{Y_{n+1} \in \mathcal{C}_{\hat{\lambda}}(X_{n+1})\right\} \geqslant 1 - \alpha.$$

Provide a counterexample to show that this need not hold if there are ties in S_i's with positive probability, even if they are exchangeable.

Solution Let $(x, y) \in \mathcal{X} \times \mathcal{Y}$ be given. If $\{\lambda_m\}_{m=1}^{\infty}$ is any decreasing sequence such that $\lambda_m > s(x, y)$ for every m and $\lambda_m \to s(x, y)$ as $m \to \infty$, then $L(y, \mathcal{C}_{\lambda_m}(x)) = 1$ for all m but $L(y, \mathcal{C}_{s(x,y)}(x)) = 0$, proving that $L(y, \mathcal{C}_\lambda(x))$ is not always right continuous.

We move to compute $\hat{\lambda}$. The definition of $\hat{R}(\lambda)$ and simple arithmetic rearrangement shows that, for any $\lambda \geqslant 0$,

$$\hat{R}(\lambda) \leqslant \alpha - \frac{\alpha - 1}{n}$$

if and only if

$$\sum_{i=1}^{n} \chi_{\{S_i < \lambda\}} \geqslant (n+1)(1-\alpha).$$

Thus, it is obvious that

$$\hat{\lambda} = S_{(\lceil (n+1)(1-\alpha) \rceil)}.$$

Here we have used order statistics $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ for the first n values of $S_1, \ldots, S_n$. Consequently:

$$C_{\hat{\lambda}}(X_{n+1}) = \left\{ y \in \mathcal{Y} \mid s(X_{n+1}, y) < S_{(\lceil (n+1)(1-\alpha) \rceil)} \right\}.$$

To prove coverage, let $\alpha \in (0,1)$ be given. Assume first that that $(S_1, \ldots, S_{n+1})$ is exchangeable and there is almost surely no tie in the S_i's. Then

$$P\left\{ Y_{n+1} \in C_{\hat{\lambda}}(X_{n+1}) \right\} = P\left\{ S_{n+1} < S_{(\lceil (n+1)(1-\alpha) \rceil)} \right\} = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geqslant 1 - \alpha,$$

where the second equality uses both assumptions: when $S_{(1)} < \cdots < S_{(n)}$, inserting a new exchangeable point S_{n+1} to get exactly right before $\lceil (n+1)(1-\alpha) \rceil$th place has the probability $\frac{\lceil (n+1)(1-\alpha) \rceil}{n+1}$. If we remove the assumption that there are no ties, then everything can go wrong. For example, just setting $s \equiv 0$. Then $\hat{\lambda} = 0$ and $C_{\hat{\lambda}}(x) = \emptyset$ for all x, clearly not yielding any good coverage probability. ////

Exercise 2

Consider split conformal prediction with data set $(X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ with given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. For $j \in \{1, 2\}$, the conformal p-value is the random variable defined by

$$p_j = \frac{1 + \sum_{i=1}^{n} \chi_{\{S_{n+j} \leqslant S_i\}}}{n+1}.$$

Given $\alpha \in (0,1)$, let $\mathcal{C}$ be the conformal interval calibrated only on the first n observations. In other words, for $j \in \{1, 2\}$, the event $\{Y_{n+j} \in \mathcal{C}(X_{n+j})\}$ equals to $\{p_j < \alpha\}$. Assuming that $((X_1, Y_1), \cdots, (X_{n+2}, Y_{n+2}))$ is exchangeable, s is a symmetric score function, and that the random vector (p_1, p_2) satisfies the positive regression dependence on subsets (PRDS) assumption given in definition 10.3 of the textbook, prove that

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}),\, Y_{n+2} \in \mathcal{C}(X_{n+2})\} \geqslant (1-\alpha)^2$$

without explicitly computing the probability on the left hand side of the inequality.

Solution First of all, Theorem 3.2 in the textbook already guarantees that, for $j \in \{1, 2\}$:

$$P\{Y_{n+j} \in \mathcal{C}(X_{n+j})\} \geqslant 1 - \alpha.$$

We focus on showing that

$$P\left\{ Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1}) \right\} \geqslant 1 - \alpha.$$

In order to make use of the PRDS condition, define a function g on the support of the discrete random variable p_1 by setting, for all t, that

$$g(t) = P\left(\{p_2 > \alpha\} \mid p_1 = t\right).$$

Note the following facts:

1. g is monotonically increasing because $g(t) = P\left(\{(p_1, p_2) \in A\} \mid p_1 = t\right)$ with the set A defined by all $(u, v) \in \mathbb{R}^2$ with $v < \alpha$, who is an increasing set.
2. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is the desired conditional probability. In symbols:

$$\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.$$

3. The expectation of $g(p_1)$ is the marginal covergae of (X_{n+2}, Y_{n+2}). In symbols:

$$\mathbb{E}\left[g(p_1)\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\}.$$

4. The average value of $g(p_1)$ on the set $\{p_1 > \alpha\}$ is greater or equal to the average value of $g(p_1)$ on the set $\{p_1 \leqslant \alpha\}$ because g is monotonically increasing. That is:

$$\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \geqslant \frac{1}{P\{p_1 \leqslant \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right]$$

The proof is complete once we combine the last three facts with the marginal coverage on (X_{n+2}, Y_{n+2}):

$$\begin{aligned}
&1 - \alpha \geqslant P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2})\right\} = \mathbb{E}\left[g(p_1)\right] \\
&= \frac{P\{p_1 > \alpha\}}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] + \frac{P\{p_1 \leqslant \alpha\}}{P\{p_1 \leqslant \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 \leqslant \alpha\}\right] \\
&\geqslant (P\{p_1 > \alpha\} + P\{p_1 \leqslant \alpha\})\frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] \\
&= \frac{1}{P\{p_1 > \alpha\}}\mathbb{E}\left[g(p_1); \{p_1 > \alpha\}\right] = P\left\{Y_{n+2} \in \mathcal{C}(X_{n+2}) \mid Y_{n+1} \in \mathcal{C}(X_{n+1})\right\}.
\end{aligned}$$

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Exercise 3

Suppose $(X_1, Y_1), \ldots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ are exchangeable. After observing X_{n+1} and X_{n+2}, select $I \in \{1, 2\}$, where I may depend on both (X_{n+1}, X_{n+2}). Given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, define a (full) conformal prediction set $\mathcal{C}(X_{n+I})$ such that

$$P\{Y_{n+I} \in \mathcal{C}(X_{n+I})\} \geqslant 1 - \alpha.$$

Solution The solution is to use a Benferroni correction for the two test points.

Specifically, put $S_i = s(X_i, Y_i)$ and let $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ be the order statistics for $S_1, \ldots, S_n$. Define

$$k = \left\lceil (n+1)\left(1 - \frac{\alpha}{2}\right)\right\rceil,$$

$$\hat{q} = \begin{cases} S_{(k)}, & k \leqslant n, \\ \infty, & k = n+1, \end{cases}$$

and

$$\mathcal{C}(X_{n+I}) = \{y \in \mathcal{Y} \mid s(X_{n+I}, y) \leqslant \hat{q}\}.$$

Standard results (for example, for full conformal prediction, Theorem 3.2 in the textbook) show that, for $i \in \{1, 2\}$, we have

$$P\{Y_{n+i} \notin \mathcal{C}(X_{n+i})\} \leqslant \frac{\alpha}{2}.$$

Thus:

$$P\{Y_{n+I} \notin \mathcal{C}(X_{n+I})\} = P\left(\bigcup_{i=1}^{2} \{Y_{n+i} \notin \mathcal{C}(X_{n+i})\}\right) \leqslant 2 \times \frac{\alpha}{2} = \alpha.$$

This completes the proof. ////

Exercise 4

Consider split conformal prediction with calibration data set $(X_1, Y_1), \cdots, (X_n, Y_n) \in \mathcal{X} \times \mathcal{Y}$ at given level $\alpha \in (0, 1)$ and given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. We write $S_i = s(X_i, Y_i)$ for all i for simplicity. We look at the aggregating prediction sets by the majority vote $\mathcal{C}^{\text{mv}}(X_{n+1})$ of two conformal algorithms $\mathcal{C}_1$ and $\mathcal{C}_2$, where $\mathcal{C}_1$ uses s as score and $\mathcal{C}_2$ uses $-s$ as score. Give an explicit formula of $\mathcal{C}^{\text{mv}}(X_{n+1})$ and its marginal coverage level, assuming exchangeability of the data set, symmetry of s, and almost surely no ties in S_i's.

Solution We first compute the expression of each conformal set. Let us use the order statistics $S_{(1)} < \cdots < S_{(n)}$ of $S_1, \cdots, S_n$. Put

$$k = \lceil (n+1)(1-\alpha) \rceil.$$

We discuss by cases of whether $k = n + 1$.

First of all, in the usual case, $k < n + 1$. The rule $(-v)_{(i)} = -v_{(n-i+1)}$ for any $v \in \mathbb{R}^n$ tells us that, by definition of $\mathcal{C}_1$ and $\mathcal{C}_2$:

$$\mathcal{C}_1(x) = \left\{y \in \mathcal{Y} \mid s(x, y) \leqslant S_{(k)}\right\}$$

and

$$\mathcal{C}_2(x) = \left\{y \in \mathcal{Y} \mid s(x, y) \geqslant S_{(n+1-k)}\right\}.$$

Thus, by definition:

$$\mathcal{C}^{\text{mv}}(x) = \left\{y \in \mathcal{Y} \mid \frac{1}{2}\sum_{i=1}^{2} \chi_{y \in C_i(x)} > \frac{1}{2}\right\} = \mathcal{C}_1(x) \cap \mathcal{C}_2(x) = \left\{y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(x, y) \leqslant S_{(k)}\right\}.$$

Next, for the trivial case $k = n + 1$, we have $\mathcal{C}^{\text{mv}}(x) = \mathcal{C}_1(x) = \mathcal{C}_2(x) = \mathcal{Y}$. Therefore:

$$\mathcal{C}^{\text{mv}}(X_{n+1}) = \begin{cases} \{y \in \mathcal{Y} \mid S_{(n+1-k)} \leqslant s(X_{n+1}, y) \leqslant S_{(k)}\} & k = \lceil (n+1)(1-\alpha) \rceil < n+1, \\ \mathcal{Y} & k = n+1. \end{cases}$$

A simple counting of the probability of placing S_{n+1} gives

$$P\left\{Y_{n+1} \in \mathcal{C}^{\text{mv}}(X_{n+1})\right\} = \frac{\max\{0, 2k-n-1\}}{n+1} = \max\left\{0, 1 - \frac{2\lfloor \alpha(n+1) \rfloor}{n+1}\right\}.$$

It just happens that this formula works for $k = n+1$ case as well. This completes the computation. ////

Chapter 3

Assignment 3 Done 2026 September 24 Due 2026 November 1

Remark All references are to our course textbook "Theoretical Foundations of Conformal Prediction" by Angelopoulos et al. (2026).

Exercise 9.1

Consider full conformal prediction with an exchangeable dataset and a symmetric score function. Instead of comparing the test score against all n training scores, compare it against a random subset indexed by $I \subseteq \{1, \ldots, n\}$, sampled independently of the data:

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathcal{Y} : S_{n+1}^{y} \leqslant \text{Quantile}\left((S_i^y)_{i \in I}; (1-\alpha)\left(1 + \frac{1}{|I|}\right)\right)\right\}.$$

Prove that this modified prediction set has marginal coverage at least $1 - \alpha$.

Solution Let $Z_i = (X_i, Y_i)$, $D_{n+1} = (Z_1, \ldots, Z_{n+1})$, and define the scores at the true test response by

$$S_i = s(Z_i; D_{n+1}), \qquad i = 1, \ldots, n+1.$$

Symmetry of s and exchangeability of the data imply that $(S_1, \ldots, S_{n+1})$ is exchangeable. Indeed, for every permutation π of the $n+1$ indices,

$$(S_{\pi(i)})_{i=1}^{n+1} = \big(s(Z_{\pi(i)}; (D_{n+1})_\pi)\big)_{i=1}^{n+1} \overset{d}{=} (S_i)_{i=1}^{n+1}.$$

The equality uses symmetry in the dataset argument, and the equality in distribution uses exchangeability of the data.

Now condition on $I = J$, for any fixed nonempty subset J having positive probability, and put $m = |J|$. Independence of I and the data means that conditioning on $I = J$ does not change the distribution of the scores. Thus the m selected scores together with S_{n+1} remain exchangeable. Let

$$k_m = \lceil (m+1)(1-\alpha) \rceil.$$

If $k_m \leqslant m$, the quantile in the question is the k_mth smallest score among $(S_i)_{i \in J}$. The exchangeable-rank argument gives

$$P\{Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid I = J\} \geqslant \frac{k_m}{m+1} \geqslant 1 - \alpha.$$

For completeness, break any score ties uniformly at random solely for this rank argument. The rank of S_{n+1} among the $m+1$ scores is then uniform on $\{1, \ldots, m+1\}$. Rank at most k_m implies that S_{n+1} is no larger than

the selected-sample quantile, which proves the first inequality even when scores tie. If $k_m = m + 1$, the quantile is $+\infty$ by the conformal convention, and the conditional coverage is 1.

Averaging the conditional guarantees over all possible subsets yields

$$\boxed{P\{Y_{n+1} \in \mathcal{C}(X_{n+1})\} = \sum_J P(I = J) P\{Y_{n+1} \in \mathcal{C}(X_{n+1}) \mid I = J\} \geqslant 1 - \alpha.}$$

If an empty subset is allowed, define its threshold to be $+\infty$ and return $\mathcal{Y}$ on that event; the same conclusion follows. Notice that I need not be uniform among subsets, nor have a fixed size. ////

Exercise 9.2

For $\mathcal{Y} = \mathbb{R}$, consider K-nearest-neighbor regression,

$$\hat{f}(x; D) = \frac{1}{K} \sum_{j \in N(x,D)} y_j,$$

where $N(x, D)$ indexes the K nearest feature vectors to x in D. Show that this algorithm has the affine dependence on the candidate response required by (9.4), so that its full conformal prediction set can be computed efficiently.

Solution Fix the training data $D_n = ((X_j, Y_j))_{j=1}^n$ and the test feature X_{n+1}. For a candidate response y, let

$$D_{n+1}^y = ((X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, y)).$$

Use a fixed rule for distance ties that depends only on features and indices, rather than response values. Then the neighbor set $N(x', D_{n+1}^y)$ does not depend on y: the augmented feature vectors are the same for every candidate. Denote this set by $N_{x'}$. Separating out the possible contribution of the test point gives

$$\begin{aligned} \hat{f}(x'; D_{n+1}^y) &= \frac{1}{K} \sum_{j \in N_{x'} \cap \{1,\ldots,n\}} Y_j + \frac{\mathbf{1}\{n+1 \in N_{x'}\}}{K}\, y \\ &= a(x'; D_n, X_{n+1}) + b(x'; D_n, X_{n+1})\, y, \end{aligned}$$

where

$$\boxed{a(x'; D_n, X_{n+1}) = \frac{1}{K} \sum_{j \in N_{x'} \cap \{1,\ldots,n\}} Y_j, \qquad b(x'; D_n, X_{n+1}) = \frac{\mathbf{1}\{n+1 \in N_{x'}\}}{K}.}$$

Both coefficients are independent of y, establishing (9.4).

To see explicitly why this helps, write $a_i = a(X_i; D_n, X_{n+1})$ and $b_i = b(X_i; D_n, X_{n+1})$. For the absolute residual score, the training and test scores are

$$S_i^y = |Y_i - a_i - b_i y| \quad (i \leqslant n), \qquad S_{n+1}^y = |(1 - b_{n+1})y - a_{n+1}|.$$

Thus

$$\mathcal{C}(X_{n+1}) = \left\{ y \in \mathbb{R} : \frac{1 + \sum_{i=1}^n \mathbf{1}\{|Y_i - a_i - b_i y| \geqslant |(1 - b_{n+1})y - a_{n+1}|\}}{n+1} > \alpha \right\}.$$

Each comparison can change only at a solution of one of the two linear equations

$$Y_i - a_i - b_i y = \pm\big((1 - b_{n+1})y - a_{n+1}\big).$$

Equations that hold identically contribute no boundary; each remaining equation has at most one solution. After computing the neighbor sets and coefficients once, there are therefore at most $2n$ candidate boundary points. Evaluate the p-value between consecutive boundary points and at the boundary points themselves to recover the exact prediction set. ////

Exercise 9.3

Fix $n \geqslant 1$. Consider the naive discretization strategy (9.6), using a finite uniform grid $G = \{y^{(1)}, \ldots, y^{(M)}\}$ over $[-B, B]$. Construct a distribution P and a symmetric score function s such that this approximation has zero marginal coverage for every choice of grid size M and range $B > 0$.

Solution First consider the nontrivial range $\alpha \geqslant 1/(n+1)$, with $0 < \alpha < 1$. The very small α case is addressed below. Take $\mathcal{X} = \mathcal{Y} = \mathbb{R}$ and define P by

$$X \sim \mathrm{Unif}(0, 1), \qquad Y = X.$$

Sample (X_i, Y_i) independently from P and choose

$$\boxed{s((x, y); D) = |y - x|.}$$

This score is symmetric because it does not depend on D at all.

For every candidate response y, all n training scores are zero:

$$S_i^y = |Y_i - X_i| = 0, \qquad i = 1, \ldots, n,$$

whereas the test score is $S_{n+1}^y = |y - X_{n+1}|$. Since $k = \lceil (n+1)(1-\alpha) \rceil \leqslant n$, the conformal quantile of the training scores is zero. Consequently, the exact full conformal prediction set is

$$\mathcal{C}_{\text{full}}(X_{n+1}) = \{X_{n+1}\},$$

which covers $Y_{n+1} = X_{n+1}$ with probability 1.

Now fix any finite grid G. Since X_{n+1} is continuously distributed,

$$P\{X_{n+1} \in G\} \leqslant \sum_{m=1}^{M} P\{X_{n+1} = y^{(m)}\} = 0.$$

Therefore, almost surely every grid point has strictly positive test score and fails the comparison against the zero calibration quantile. The accepted grid-index set is empty. In particular, the expansion of accepted grid points into intervals in (9.6) produces

$$\mathcal{C}_{\text{naive}}(X_{n+1}) = \bigcup_{m \in \varnothing} (y^{(m-1)}, y^{(m+1)}) = \varnothing \quad \text{almost surely.}$$

Hence

$$\boxed{P\{Y_{n+1} \in \mathcal{C}_{\text{naive}}(X_{n+1})\} = 0.}$$

The same P and s work for every fixed finite grid, including every uniform grid of any resolution or range. For $M = 1$, interpret the grid as any chosen single point; the argument is unchanged.

The necessary restriction on α As written, the exercise omits a restriction needed for its conclusion. If $0 < \alpha < 1/(n+1)$, then

$$\lceil (n+1)(1-\alpha) \rceil = n + 1.$$

The conformal quantile is then $+\infty$, so every grid point is accepted for any real-valued score function. With the boundary convention $y^{(0)} = -\infty$ and $y^{(M+1)} = +\infty$ from Section 9.2.3,

$$\mathcal{C}_{\text{naive}}(X_{n+1}) = \bigcup_{m=1}^{M} (y^{(m-1)}, y^{(m+1)}) = \mathbb{R}.$$

Its coverage is 1, making zero coverage impossible in this range. Thus the counterexample applies precisely when $\alpha \geqslant 1/(n+1)$, or equivalently when the conformal threshold is a finite calibration order statistic. ////

Exercise 9.4

Let A be a regression algorithm that takes an ordered dataset D of size m and returns $\hat{f}(\cdot; D)$, without assuming symmetry. Let $\xi \sim \mathrm{Unif}[0,1]$ and choose a map $\xi \mapsto \sigma_\xi \in \mathfrak{S}_m$ such that σ_ξ is uniform among all permutations. Define

$$s((x,y); D, \xi) = |y - \hat{f}(x; D_{\sigma_\xi})|.$$

Prove that s is a symmetric randomized score function in the sense of Definition 9.1.

Solution Fix any dataset $D = (z_1, \ldots, z_m)$ and any permutation $\pi \in \mathfrak{S}_m$, where $z_j = (x_j, y_j)$. If $\sigma \in \mathfrak{S}_m$, then

$$(D_\pi)_\sigma = (z_{\pi(\sigma(1))}, \ldots, z_{\pi(\sigma(m))}) = D_{\pi\circ\sigma}.$$

The map $\sigma \mapsto \pi\circ\sigma$ is a bijection of $\mathfrak{S}_m$, with inverse $\rho \mapsto \pi^{-1}\circ\rho$. Thus composing a uniform random permutation with the fixed permutation π preserves its uniform distribution:

$$\pi \circ \sigma_\xi \overset{d}{=} \sigma_\xi.$$

In particular, shuffling D_π uniformly has the same distribution as shuffling D uniformly.

To verify the full requirement of Definition 9.1, fix any finite list of evaluation points $(x'_1, y'_1), \ldots, (x'_r, y'_r)$ and define

$$F(\rho) = \big(|y'_j - \hat{f}(x'_j; D_\rho)|\big)_{j=1}^r, \qquad \rho \in \mathfrak{S}_m.$$

Using the same random seed for all r scores, we obtain

$$\begin{aligned}\big(s((x'_j, y'_j); D_\pi, \xi)\big)_{j=1}^r &= F(\pi \circ \sigma_\xi) \\ &\overset{d}{=} F(\sigma_\xi) \\ &= \big(s((x'_j, y'_j); D, \xi)\big)_{j=1}^r.\end{aligned}$$

This proves equality in distribution of the entire score vector for arbitrary evaluation points and any permutation of D, exactly as required by Definition 9.1. No symmetry of A is needed: the random shuffle gives every possible ordering of the data the same probability. ////